

Stockholm Public Transport Delay Analytics Using Apache Spark

Group 67: Hieu Vu, Rong Jin, Serban

1. Problem Description

Stockholm's public transportation systems generate large amounts of operational data, identifying delay patterns across routes, stations, and time periods that require scalable data processing.

This project investigates delay patterns in Stockholm's public transport network by analyzing GTFS (General Transit Feed Specification) data and real-time delay data (where available). The goal is to identify which routes and stations experience the highest delays, how delays vary throughout the day, and whether weekday and weekend patterns differ.

We aim to answer questions such as:

- Which transport lines have the highest average delays?
- Which stations experience the most frequent delays?
- During which hours are delays most severe?
- How do delay patterns differ between weekdays and weekends?

2. Tools

The project will use a distributed data processing pipeline:

- Storage: HDFS
- Processing: Apache Spark (PySpark)
- Data Format: CSV and GTFS files converted to Parquet
- Visualization: Plotly Dash (or Power BI)
- Development: Docker, Jupyter Notebook, Git, Python

3. Data

We will use Stockholm's public transport open dataset provided by SL. We will create a free API key from Trafiklab (create a free account on Trafiklab, create a project, enable the GTFS Regional dataset) and download the dataset using the key

(e.g.: https://opendata.samtrafiken.se/gtfs/sl/sl.zip?key=YOUR_API_KEY)

The dataset includes GTFS files containing: *stops.txt*, *routes.txt*, *trips.txt*, *stop_times.txt*, *calendar.txt*.

4. Methodology and Algorithm

Our pipeline contains 4 stages:

- Data ingestion: Download GTFS dataset, upload files into HDFS
- Data preprocessing: Load data into Spark DF, clean missing values, join routes, trips, stops, and stop times
- Distributed analytics: calculate average delays per route and station, aggregate delays by hour of day, compare weekday versus weekend patterns, rank the busiest stations.
- Visualization: Create charts showing delay distributions, display hours trends, present a map highlighting stations with the largest delay.